

LED Matrix Display Using a Single LM3914

Concept

Using ten **LM3914** ICs is not space-efficient or cost-effective. Instead, a multiplexing technique can be employed so that a **single LM3914** performs the same function.

By rapidly switching between channels, the analyzer can be extended to support up to **32 input channels** while significantly reducing hardware cost and PCB area.

System Overview

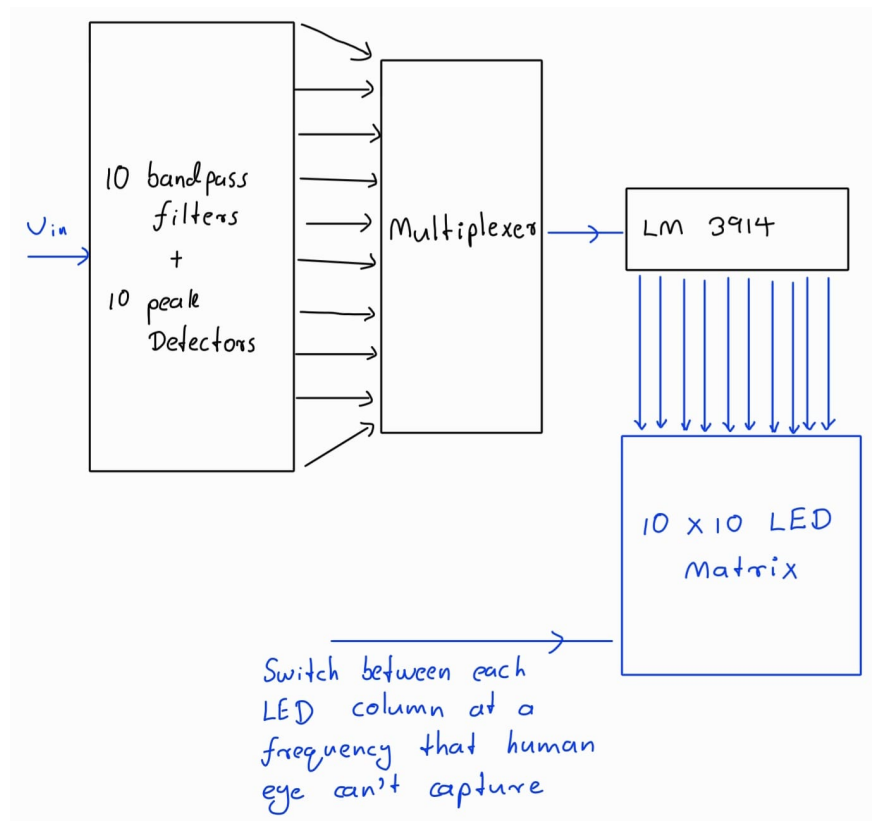


Figure 1: System for use single LM3914

Operating Principle

- The input signal is divided into ten frequency bands using band-pass filters.

- Each filter output is processed by a peak detector.
- The ten peak voltages are sequentially selected by a multiplexer.
- The selected signal is fed into a single LM3914.
- The LM3914 drives one column of the LED matrix at a time.
- By rapidly scanning all columns, persistence of vision makes the display appear continuously illuminated.

Advantages

- Lower component cost
- Reduced PCB area
- Easier routing
- Scalable architecture (up to 32 channels with additional multiplexing)
- Lower power consumption compared to using one LM3914 per channel